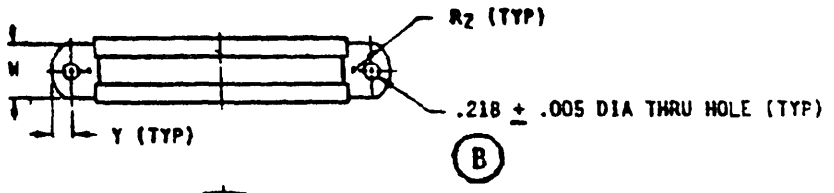
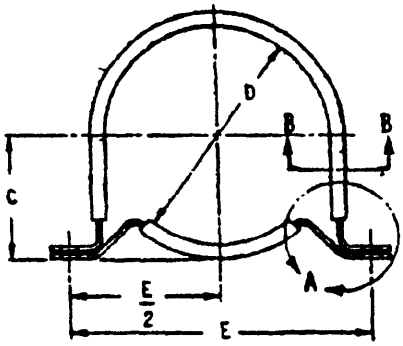
MILITARY SPECIFICATION SHEET

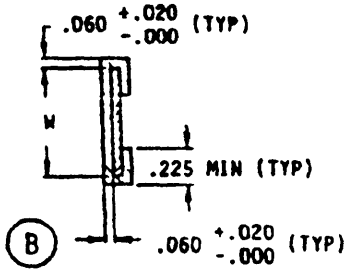
CLAMP ASSEMBLY, SADDLE-TYPE, CUSHIONED, SILICONE FABRIC, REINFORCED 321 CRES, 500°F, GENERAL PURPOSE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

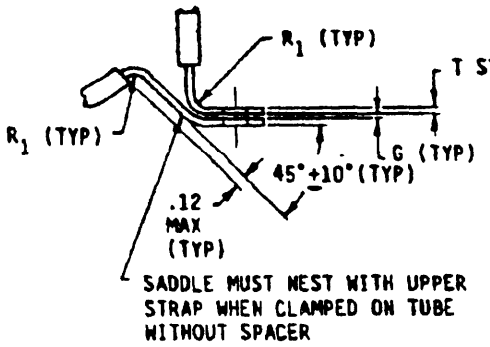
The complete requirements for procuring the clamps described herein shall consist of this document and the latest issue in effect of specification MIL-C-85449.







SECTION B-B



DETAIL A

TOLERANCES: .XXX ± .010
.XX ± .03
UNLESS OTHERWISE SPECIFIED

AMSC N/A

FSC 5340

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

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TABLE I. CLAMP DIMENSIONS

DASH NO.	TUBE SIZE	C ± .020	D (5)	E ±.015 (5)	G ±.001 (7) (B)	R_1 (B)	R_2	T (6)	W +.015 -.000	Y	WEIGHT, LBS. MAX.
-10	5/8	0.493	.625	1.525	.020	.090	.37	.040	.500	.218	0.0443
-12	3/4	0.555	.750	1.650							0.0490
-14	7/8	0.618	.875	1.775							0.0537
-16	1	0.690	1.000	1.958	.030	.109	.50	.050	.625	.250	0.0856
-18	1 1/8	0.753	1.125	2.083							0.0923
-20	1 1/4	0.815	1.250	2.208							0.0989
-24	1 1/2	0.940	1.500	2.458							0.1120
-28	1 3/4	1.065	1.750	2.708							0.1252
-32	2	1.203	2.000	3.016		.125		.063	.750		0.1961
-36	2 1/4	1.328	2.250	3.266							0.2143
-40	2 1/2	1.453	2.500	3.516							0.2324
-44	2 3/4	1.578	2.750	3.766							0.2505
-48	3	1.703	3.000	4.016							0.2687

REQUIREMENTS:

(B) Materials: Strap and Saddle - 321 CRES sheet per AMS 5510, except that a tensile strength of 100-120 KSI shall be required. Strip to have edges rolled to a radius of T/4.

Cushion - Fabric reinforced, silicone rubber, 65-75 durometer, color light blue, per the general specification with additional requirements listed in Table II of this specification sheet. The fabric reinforcing material shall be molded into the band side of the clamp cushion at 45 angle (See Figure A).

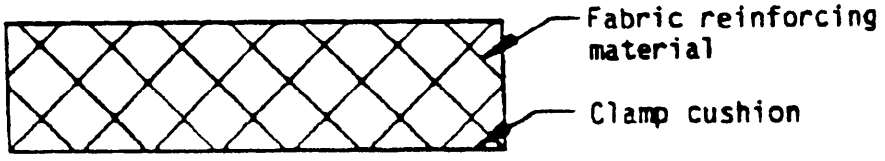


FIGURE A

(B) Finish: Strap and saddle, ball burnish to round edges to a .010 radius. Cleaned and passivated per QQ-P-35 latest revision.

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Marking: Impression stamp complete part number and manufacturer's trademark on strap, visible with cushion in place above the center line. The saddle shall have the dash size impression stamped in the vicinity of the mounting hole away from any bend radii.

Part No: Part numbers shall consist of the following: (In sequence)

1. The letter "M"
2. The basic specification number
3. A slash and slash number of this specification sheet
4. A dash and the appropriate size dash numbers from Table I

Example:

(B) M85449 /3 - 16

- 16 — dash number (-16 = 1" Tube Size)
- /3 — slash sheet number
- M85449 — basic military specification number

CUSHION REQUIREMENTS:

1. Physical properties - as specified in Table II. Unless otherwise specified, results are an average of 5 specimens and tolerance of temperature is ± 5°F.

TABLE II. CUSHION MATERIAL PHYSICAL PROPERTIES

TEST	TEST METHOD	REQUIRED ORIGINAL PROPERTIES	ALLOWABLE CHANGE FROM ACTUAL ORIGINAL PROPERTIES AFTER:		
			HEAT AGING 70 HRS. AT +500°F	OIL IMMERSION IN MIL-L-7808 70 HRS. AT +302°F	DIP TEST *
HARDNESS, DUROMETER "A"	ASTM D2240	65 - 75	+10 PTS MAX.	-30 PTS MAX.	+5 PTS
TENSILE STRENGTH - PSI	**ASTM D412	1200 MIN.	1000 MIN.	300 MIN.	300 MIN.
ELONGATION - %	**ASTM D412	30 MIN.	30 MIN.	30 MIN.	30 MIN.
TEAR STRENGTH - PPI	ASTM D624 (DIE "B")	300 MIN.	300 MIN.	300 MIN.	300 MIN.
VOLUME CHANGE - %	ASTM D471	----------	----------	+45% MAX.	±5% MAX.
WEIGHT CHANGE - %	ASTM D471	REPORT	-5% MAX.	----------	----------
SPECIFIC GRAVITY	ASTM D471	REPORT	----------	----------	----------

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* Dip Test - Test samples shall be dipped in P-D-680 solvent and allowed to dry at room temperature for 24 hours. This procedure shall be repeated 10 times.

** Tests shall be conducted in accordance with ASTM D412 on the fabric reinforced cushion test specimen. The test specimen shall be cut from the actual molded cushion, with the cushion unfolded to provide a suitable die A specimen. The tensile strength and percent elongation shall be measured when the fabric reinforcing material begins to separate from the elastomer. Tensile strength will be based on overall thickness of rubber and fabric reinforcing material. Specimen failures due to fabric separation at the test fixture jaws shall be disregarded and another sample tested.

2. Compression Set - See MIL-C-85449
   - Air age at 302°F
   - Not to exceed 30% average of 3 specimens

3. Flammability: - See MIL-C-85449
   - Specimens: -16 size clamp assemblies unbent until flat
   - Vertical burn test

NOTES:

1. Dimensions in inches unless otherwise specified.
2. Metal parts shall be free of burrs, sharp edges, tool marks and die marks.
3. With clamp assembled on tube, no gap is permitted between cushion and tube for upper half of strap and cushioned length of saddle.
4. With cushion pushed to either end of strap, sufficient cushion shall be available to prevent strap contacting tube.

5 The "D" & "E" dimensions will be measured with mandrel clamped and a "G" thickness spacer between the clamp feet.

B 6 Tolerances per AMS 2242.

7 "G" dimension is for thickness of shim required to inspect assembled clamp.

B 8. In the event of a conflict between the text of this specification sheet and the references cited herein, the text of this specification sheet shall take precedence.

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INTENDED USE:

These clamps are intended for use as follows:

Temperature Range:

-65 F to +500 F. These clamps have cushions which have been compounded for high temperature capability at the sacrifice of fluid resistance. Use is recommended only in those areas with operating temperatures from 275 F to 500 F where fluid exposure is minimal. For application below 275 F the use of clamps with a higher degree of fluid resistance is recommended such as MIL-C-85449/1 or /2 clamps.

Systems:

1. All fuel and electrical system applications in high temperature areas.

2. In electrical systems for support of wire bundles provided multiple wire bundles are wrapped or contained in conduits or other wire protection tubing.

3. In systems with high vibration areas such as near pumps, motors etc. It is to be noted that MIL-C-85449 requires tests only up to the -24 size clamp. Larger clamps used in high vibration areas should be subjected to additional testing.

Custodians:
Navy - AS
Air Force - 99
Army - AV

Reviewer Activities:
Air Force - 82
DLA - IS

Preparing Activity:
Navy - AS

Project No.: 5340-1833

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